

Digital Corporate Law

Lecture 1: Why Regulate AI?

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Course logistics

- 1 credit course
- Around 15 lectures total
 - 1 mark for attending one lecture
 - Maximum of 10 attendance marks
- 90 marks project
 - To be done in a group of two
 - Write a report and submit by a deadline (yet to be announced)

Audience for the course

- CS graduates
 - Responsible for making AI applications
- Lawyers
 - Improving legal certainty

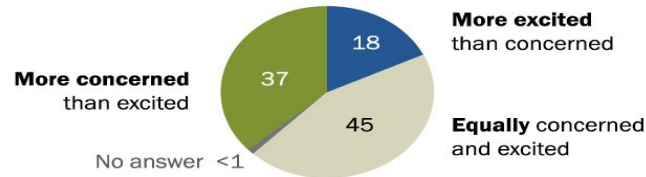
Topics to be covered

- Why regulate AI?
- What is AI?
 - Challenges in legal definition
 - How various jurisdictions address this
- Fundamental rights impact assessment
- Specifics of the EU AI Act
 - First comprehensive law on regulating AI
 - Subject matter and scope
 - Terminology
 - Risk pyramid
 - Prohibited, high risk AI
 - Requirements for high risk AI systems
 - Obligations of providers and deployers of high risk AI systems and other parties
 - Transparency obligations
 - Obligations for providers of GPAI models
 - Measures in support of innovation
- Ongoing lawsuits against the tech giants

Why so much hype over regulating AI?

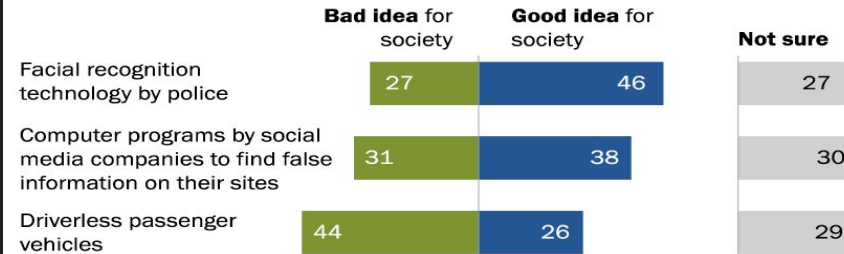
Americans lean toward concern over excitement when it comes to the increased use of AI in daily life ...

% of U.S. adults who say that overall, the increased use of artificial intelligence computer programs in daily life makes them feel ...



And public views are varied when it comes to three specific AI applications

% of U.S. adults who think widespread use of each of the following has been/would be a ...



Note: Respondents who did not give an answer are not shown. Figures may not add up to 100% due to rounding.

What is new with AI?

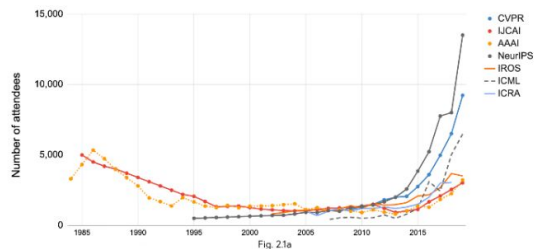
- Computers have been there for decades now
 - Every hazard listed has been there
- The important difference is

SCALE

And Exponential Growth

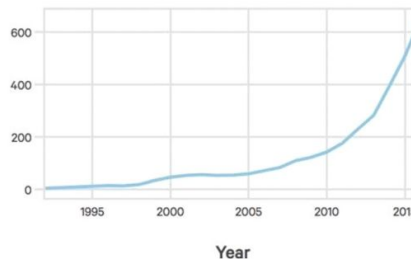
- The speed at which AI is evolving gives too little time for the society to respond
 - Implying legal backwardness

Attendance at large conferences (1984-2019)
Source: Conference provided data.



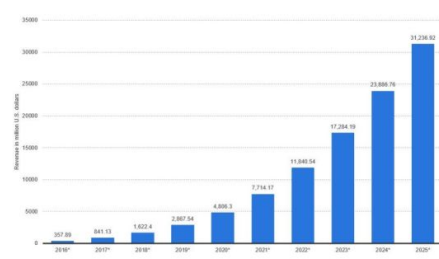
**Number of attendance
At AI conferences**

Source: The Gradient



**Startups Developing AI
Systems**

Source: Crunchbase, VentureSource, Sand
Hill Econometrics



**Enterprise Application AI
Revenue**

Source: Statista

Potential to Impact Democracy!

- Possible to generate huge amount of misinformation
 - at virtually no cost!
- An era in which nobody trusts anything!

AI-generated disinformation poses threat of misleading voters in 2024 election

Politics May 14, 2023 7:52 PM EST

WASHINGTON (AP) — Computer engineers and tech-inclined political

A.I. Muddies Israel-Hamas War in Unexpected Way

Fakes related to the conflict have been limited and largely unconvincing, but their presence has people doubting real evidence.

<https://www.pbs.org/newshour/politics/ai-generated-disinformation-poses-threat-of-misleading-voters-in-2024-election>

<https://www.nytimes.com/2023/10/28/business/media/ai-muddies-israel-hamas-war-in-unexpected-way.html>

Threatens Rule of Law!

- Denial of genuine evidences in a court of law

WATERLOO REGION

AI threatens courts with fake evidence, UW prof says

Ethics prof supports pause in development and deployment of AI until safeguards are in place



By Terry Pender Record Reporter

Tuesday, March 28, 2023 | 2 min to read

Negative Impact on Health

- AI chatbots like ChatGPT giving wrong medical advice

AI Shocker: Man Dies By Suicide After Six-Day Chatting With ChatGPT-like Chatbot

Reports claimed that the man referred to as Pierre took his own life after talking to AI bot ELIZA for six weeks on climate change. However, the conversation about the climate issues of the world turned 'confusing and harmful', prompting him to end his own life.

Times Now Bureau, Surabhi Shaurya | Updated Mar 31, 2023, 06:13 PM IST



WORLD

AI bots like ChatGPT can now influence people in taking decisions related to life and death

Experts are calling for future bots to be banned from giving advice on ethical issues, after researchers found that AI bots can seriously influence people and the decisions they take even in life and death situations, in a haphazard manner.

Mehul Reuben Das | April 07, 2023 10:36:41 IST



<https://www.timesnownews.com/technology-science/ai-shocker-man-dies-by-suicide-after-six-day-chatting-with-chatgpt-like-chatbot-article-99146570>
<https://www.firstpost.com/world/ai-bots-like-chatgpt-can-influence-people-in-decisions-related-to-life-and-death-12421072.html>

Violation of fundamental rights

- E.g. right to equality

Artificial Intelligence and Gender Equality: Is AI a Friend or Foe of Inclusion?



Alessandro Mascellino
Technology Journalist 
Last updated: 26 October, 2023

 Disclosure

 Why Trust Us





KEY TAKEAWAYS

Uncover AI's gender gap bridging potential by erasing biases in recruitment and pay, amid bias encoding challenges. Explore inclusive skill and leadership initiatives, addressing ethical concerns for a diverse, equitable tech-driven future.

<https://www.techopedia.com/artificial-intelligence-and-gender-equality-is-ai-a-friend-or-foe-of-inclusion>

Hazardous to the Environment

- OpenAI has revealed that it cost them \$100 million and took 100 days, utilizing 25,000 NVIDIA A100 GPUs. Servers with these GPUs use about 6.5 kW each, resulting in an estimated 50 GWh of energy usage during training
 - The estimated energy consumption of a GPT-4 query is 0.001-0.01 kWh (3.6-36 kJ)
- Increased e-waste!

<https://www.ri.se/en/news/blog/generative-ai-does-not-run-on-thin-air>

<https://lifestyle.livemint.com/news/big-story/ai-carbon-footprint-openai-chatgpt-water-google-microsoft-111697802189371.html>

Automation Bias

- If computers have done it, it must be right
- If AI has done it, it must be right
 - However, AI is still emerging!

Breast radiologists impacted by automation bias when using AI

Amerigo Allegretto

May 2, 2023

More examples

Pentagon “Victimized”

Artificial Intelligence
(AI)

Abené Clayton

Mon 22 May 2023 23:42 BST



Fake AI-generated image of explosion near Pentagon spreads on social media

Picture shared on verified accounts fuels concerns over AI's potential to generate misinformation



📷 Distribution of an AI-generated image briefly shook the stock market. Photograph: Patrick Semansky/AP

An AI-generated image that appeared to show an explosion next to a building in the Pentagon complex circulated on social media platforms on Monday, in the latest incident to highlight concerns over misinformation generated by AI.

The image of a tall, dark gray plume of smoke quickly spread on Twitter

← Thread



The Kobeissi Letter 🟡

@KobeissiLetter

...

This morning, an AI generated image of an explosion at the US Pentagon surfaced.

With multiple news sources reporting it as real, the S&P 500 fell 30 points in minutes.

This resulted in a \$500 billion market cap swing on a fake image.

It then rebounded once the image was confirmed fake.

AI is becoming dangerous.



🔔 Stay informed

Manipulated media [Find out more](#)



8:39 PM · May 22, 2023 · 3.7M Views

Impersonation

Chinese Man Nearly Loses \$600,000 To AI-Driven Scam In 10 Minutes



Bharat Sharma

Updated on May 26, 2023, 15:14 IST · 2 min read



“It appears that the incident took place on April 20 when the man, surnamed Guo, got a WeChat video call from someone who he thought was a friend. By using an artificial intelligence programme, a scammer was able to mimic the face and the voice of Guo's friend.”

Manipulation that can impact masses

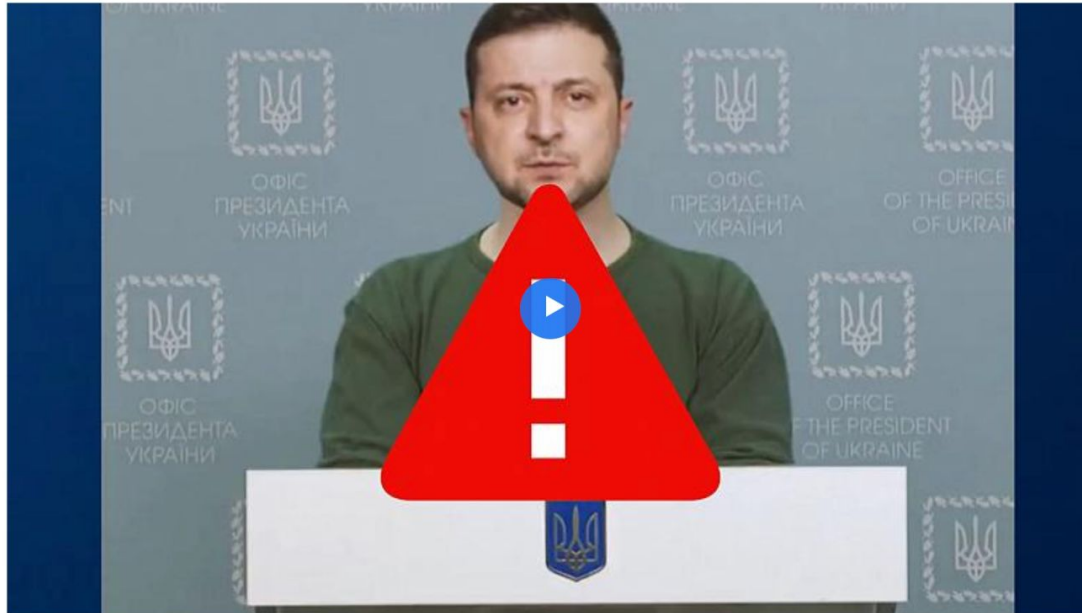


<https://www.youtube.com/watch?v=cQ54GDm1eL0>

Deepfake Zelenskyy surrender video is the 'first intentionally used' in Ukraine war COMMENTS



By **Matthew Holroyd** & **Fola Olorunselu** • Updated: 16/03/2022

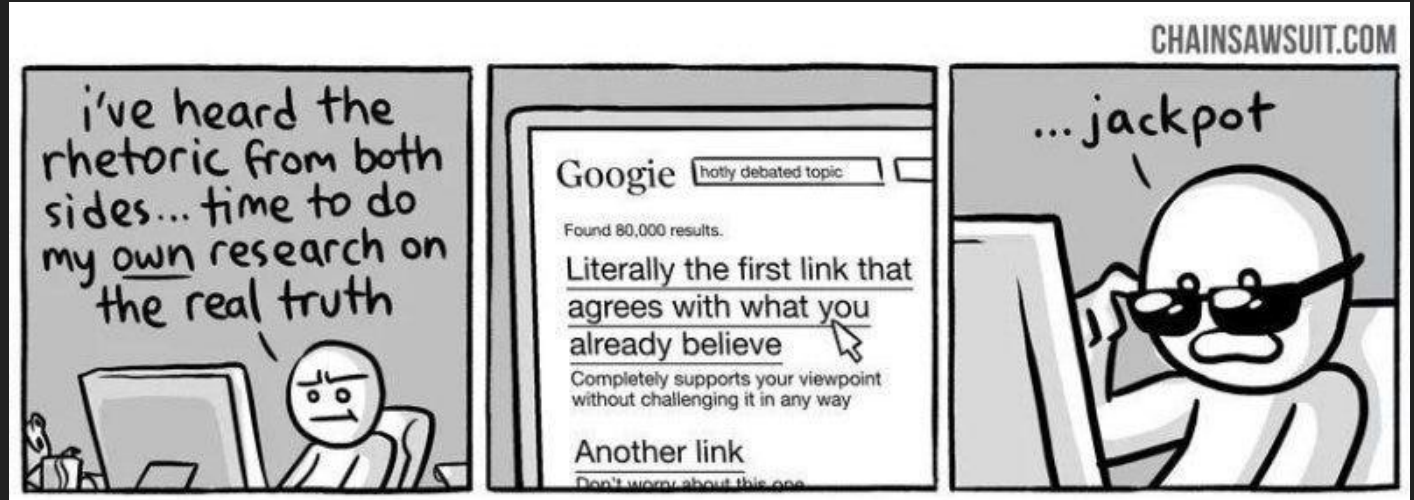


[EuroNews Article](#)
[Youtube video](#)

The deepfake video was shared during a hack on Ukrainian television. - Copyright Euronews via Twitter

<https://www.youtube.com/watch?v=X17yrEV5sl4>

Cognitive bias



Media and News!

ट्यूशन पढ़ते-पढ़ते बबलू भैया बने बाबू, 54 के बूढ़े टीचर से जवान छात्रा ने की शादी! कहा- मैं इतनी सुंदर हूं...



लोगों को हैरान कर रही है ये अजबगजब शादी (इमेज- इंस्टाग्राम)

Fake Or Real: सोशल मीडिया पर एक बेहद अजीबोगरीब स्टोरी वायरल हो रही है. इसमें एक पच्चीस साल की लड़की को अपने 54 साल के पति के साथ देखा गया. इनकी लव स्टोरी लोगों को हैरान कर रही है.

• **NEWS18 हिंदी**

• **LAST UPDATED :** FEBRUARY 2, 2024, 10:40 IST



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Channel



WRITTEN BY : **Sandhya Kumari**

They don't know the truth themselves! Whom to trust?

फेक हो सकता है मामला

सोशल मीडिया पर इस अजीबोगरीब शादी का वीडियो तेजी से वायरल हो रहा है. जहां यूट्यूब पर इस वीडियो को बिहार का बताया जा रहा है, वहीं इंस्टाग्राम पर इसे उत्तर प्रदेश के जौनपुर का मामला बताकर शेयर किया गया है. साथ ही वीडियो में कहीं भी कपल की पहचान का जिक्र नहीं किया गया है. बता दें कि इन दिनों अपने कंटेंट को वायरल करने के लिए लोग इस तरह की शादियों की कहानी बुनकर वीडियो शेयर कर रहे हैं. ताकि उनके वीडियो पर ज्यादा से ज्यादा व्यूज आए. सोशल मीडिया पर वायरल हो रही इस लव स्टोरी को भी लोग स्क्रीटेड बता रहे हैं.

नोट- वायरल कंटेंट के आधार पर खबर को बनाया गया है. न्यूज18 इसकी पुष्टि नहीं करता.

Bias!



The Switch
Wanted: The 'perfect babysitter.' Must pass AI scan for respect and attitude.



THE NEW YORK TIMES
If you're not a white male, artificial intelligence's use in healthcare could be dangerous



Women less likely to be shown ads for high-paid jobs on Google, study shows

Automated testing and analysis of company's advertising system reveals male job seekers are shown far more adverts for high-paying executive jobs



How Facebook Is Giving Sex Discrimination in Employment Ads a New Life

By Galen Sheehy, ACLU Women's Rights Project
SEPTEMBER 16, 2016 10:00 AM



DALL.E 2 Generated images of lawyers



Source: A new AI draws delightful and not-so-delightful images

<https://www.vox.com/future-perfect/23023538/ai-dalle-2-openai-bias-gpt-3-incentives>

DALL.E 2 Generated images of flight attendants



Source: A new AI draws delightful and not-so-delightful images
<https://www.vox.com/future-perfect/23023538/ai-dalle-2-openai-bias-gpt-3-incentives>

AI in recruitment and selection

https://www.theregister.com/2020/12/08/texas_compsci_phd_ai/

The Register®

OFF-PREM ▼ ON-PREM ▼ SOFTWARE ▼ SECURITY OFF-BEAT ▼ VENDOR VOICE ▼



{* ARTIFICIAL INTELLIGENCE *}

Uni revealed it killed off its PhD-applicant screening AI – just as its inventors gave a lecture about the tech

Fears of bias put compsci dept into damage-limitation mode after years of using it to analyze applications

Katyanna Quach Tue 8 Dec 2020 // 12:04 UTC

SHARE

A university announced it had ditched its machine-learning tool, used to filter thousands of PhD applications, right as the software's creators were giving a talk about the code and drawing public criticism.

// MOST READ



Apple fires warning shot at Facebook and Google on privacy, pledges fight

Predictive policing poses discrimination risk, thinktank warns

Machine-learning algorithms could replicate or amplify bias on race, sexuality and age



▲ One officer said human biases including more stop and searches of black men were likely to be introduced into algorithm data sets. Photograph: Carl Court/Getty Images

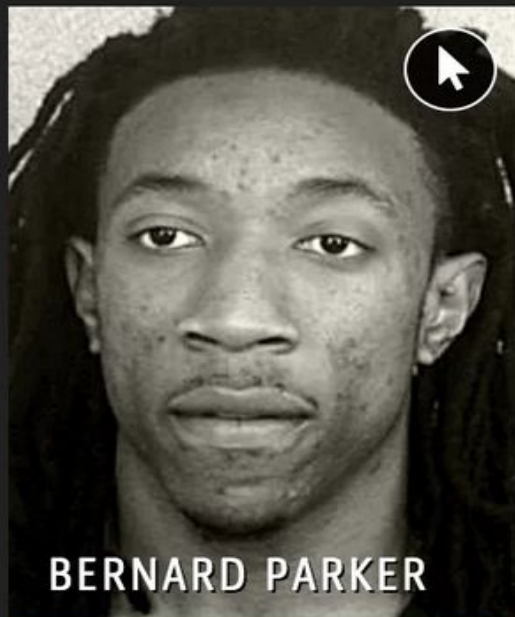
Two Drug Possession Arrests



DYLAN FUGETT

LOW RISK

3



BERNARD PARKER

HIGH RISK

10

Fugett was rated low risk after being arrested with cocaine and marijuana. He was arrested three times on drug charges after that.

Already voices against the use of AI!

Why Hundreds of Mathematicians Are Boycotting Predictive Policing

Some academics are calling the controversial practice a "scientific veneer for racism."



// BY [COURTNEY LINDER](#) JUL 20, 2020

<https://www.popularmechanics.com/science/math/a32957375/mathematicians-boycott-predictive-policing/>

Police doesn't believe it either!

Los Angeles Times



LAPD pioneered predicting crime with data. Many police don't think it works



Tattoo shop owner Edward Everett shows LAPD Senior Lead Officers Denise Vasquez, center, and Oscar Bocanegra where cars have been burglarized on Sherman Way in Reseda. The officers patrol where a computer program predicts property crimes will occur. (Mel Melcon / Los Angeles Times)

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Supreme Court of India's vision to use AI

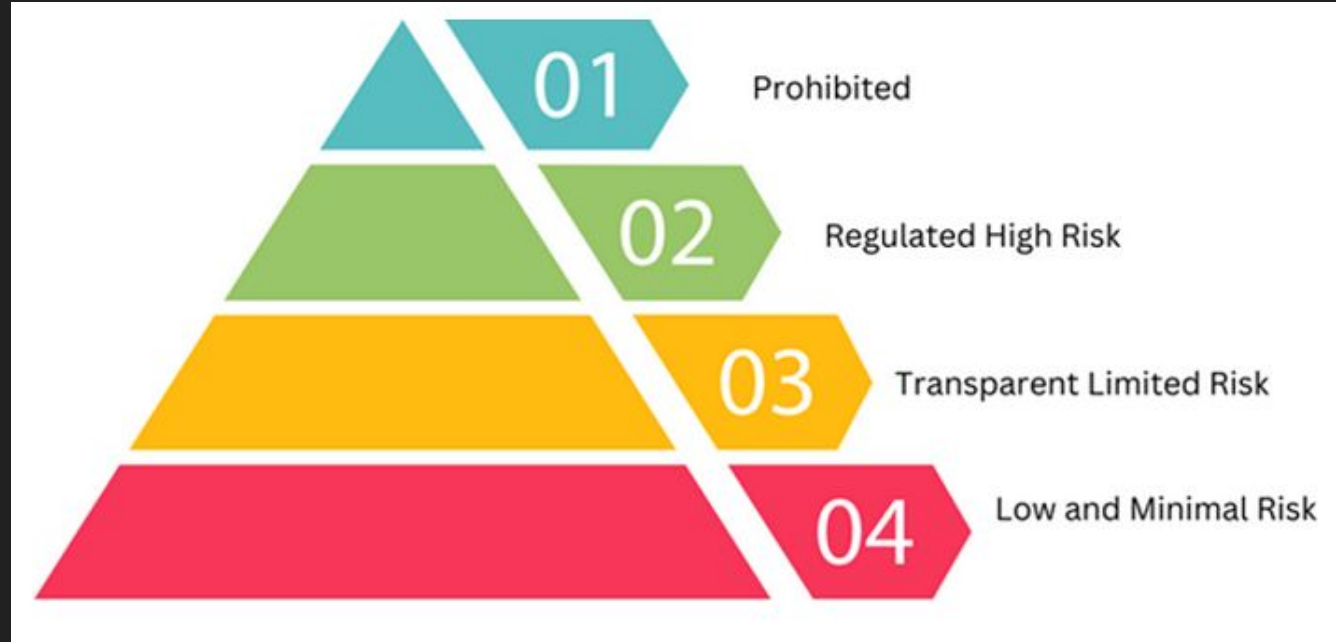
- Transcription and Translation
 - Supreme Court Constitutional Benches
- Impacts rule of law
 - Binding on all courts in India, including the Supreme Court benches with smaller strength
- Anil Ambani's case
 - Missing just one word can be catastrophic

Translation of Judgments to Indian Languages!

Language	Class
Assamese	1
Bengali	3
Gujarati	1
Hindi	4
Kannada	1
Kashmiri	1
Konkani	2
Malayalam	1
Manipuri	1
Marathi	2
Nepali	1
Oriya	1
Punjabi	2
Sanskrit	2
Sindhi	1
Tamil	3
Telugu	1
Urdu	3
Bodo	0
Santhali	1
Maithili	1
Dogri	0

Table 1: Class of 22 languages recognized by the Eighth Schedule of the Constitution of India as per Joshi et al. (2020). Higher class means higher availability of data that may enable a high quality processing.

EU AI Act



Regulating AI

Lecture 2: What is AI?

Dr. Kshitiz Verma
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Various legal instruments

- [Universal Guidelines for AI \(2018\)](#)
- [OECD AI Principles \(2019\)](#)
- [G20 AI Guidelines \(2019\)](#)
- [UNESCO Recommendation on AI Ethics \(2021\)](#)
- [EU AI Act \(Proposed\)](#)
- [Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence](#)
- [Africa CPHR - Resolution 473](#)
- [China Regulation of Recommendation Algorithms \(2022\)](#)
- [EU Digital Services Act \(Proposed, provisions on Recommendation Algorithms\)](#)

Define AI

- <https://www.linkedin.com/pulse/8-definitions-artificial-intelligence-murat-durmus-cvlje/>
- <https://www.morganlewis.com/pubs/2023/09/eu-ai-act-institutions-debate-definition-of-ai>

Recital (6)

- The notion of AI system in this Regulation should be clearly defined and closely aligned with the work of international organisations working on artificial intelligence to ensure legal certainty, facilitate international convergence and wide acceptance, while providing the flexibility to accommodate the rapid technological developments in this field.

Recital (6)

- Moreover, it should be based on key characteristics of artificial intelligence systems, that distinguish it from simpler traditional software systems or programming approaches and should not cover systems that are based on the rules defined solely by natural persons to automatically execute operations.

Recital (6)

- A key characteristic of AI systems is their capability to infer. This inference refers to the process of obtaining the outputs, such as predictions, content, recommendations, or decisions, which can influence physical and virtual environments and to a capability of AI systems to derive models and/or algorithms from inputs/data. The techniques that enable inference while building an AI system include machine learning approaches that learn from data how to achieve certain objectives; and logic- and knowledge-based approaches that infer from encoded knowledge or symbolic representation of the task to be solved. The capacity of an AI system to infer goes beyond basic data processing, enable learning, reasoning or modelling.

Recital (6)

- The term “machine-based” refers to the fact that AI systems run on machines.

Recital (6)

- The reference to explicit or implicit objectives underscores that AI systems can operate according to explicit defined objectives or to implicit objectives. The objectives of the AI system may be different from the intended purpose of the AI system in a specific context.

Recital (6)

- For the purposes of this Regulation, environments should be understood as the contexts in which the AI systems operate, whereas outputs generated by the AI system, reflect different functions performed by AI systems and include predictions, content, recommendations or decisions.

Recital (6)

- AI systems are designed to operate with varying levels of autonomy, meaning that they have some degree of independence of actions from human involvement and of capabilities to operate without human intervention.

Recital (6)

- The adaptiveness that an AI system could exhibit after deployment, refers to self-learning capabilities, allowing the system to change while in use. AI systems can be used on a stand-alone basis or as a component of a product, irrespective of whether the system is physically integrated into the product (embedded) or serve the functionality of the product without being integrated therein (non-embedded).

My take on the definition

- I think that the definition in first draft was better
- Too much emphasis on generalizing the definition
- If technology changes, just change the definition
 - Rather than making a reasonably future proof definition
- Word computing is missing
 - From the definition
 - As well as from the recital
 - I think it's a costly omission